

■ B.7.10 Saving the State of *Mathematica*

On many computer systems, you can save the complete state of your *Mathematica* session, so you can start it up again exactly where you left off.

The basic mechanism for this is to save a “core image” file, and then to execute this core image when you want to start again.

<code>Dump["file"]</code>	write out the core image of a <i>Mathematica</i> session
<code>Dump["file", start]</code>	insert the expression <i>start</i> in the core image, so the expression is evaluated when the image starts up
<code>Dump["file", start, "code"]</code>	take the file <i>code</i> , rather than <code>\$CommandLine[[1]]</code> , to be the original <i>Mathematica</i> executable

Saving the state of a *Mathematica* session.

A common way to use `Dump` is in creating copies of *Mathematica* that have certain data “pre-loaded”. If you want these copies to start up like a standard *Mathematica* session, you can use `Dump["file", $Line=0; <<init.m]`.

You should realize that on most systems `Dump` creates a file that is more than two megabytes in size. If you want to save only some aspects of your *Mathematica* session, it is much more economical of storage space to save what you need in standard ASCII form using `Save`, rather than saving a complete core image using `Dump`.

If you have files open in *Mathematica* when you call `Dump`, these files will be closed. There would be no point in saving the file pointers in the core image, since there is no guarantee that the files will still exist, or be in the same place, when the core image is restarted.